ENGLISH PAPER SUMMARY | ALL ORIGINAL NUMBERED FIGURES

WIFF1.0: a hybrid machine-learning-based parameterization of wave-induced sea ice floe fracture

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Geoscientific Model Development 15, 803-814 (2022). DOI: 10.5194/gmd-15-803-2022.

A classifier plus a neural histogram model accelerates a stochastic numerical fracture parameterization while retaining similar statistics in the tested CICE simulations. This demonstrates model emulation, not independent observational validation of fracture physics.

Prepared independently from the original paper. This is a new English synthesis, not a translation of the Korean notes. The figure atlas reproduces all 5 numbered figures with explanatory reading notes. Section and equation references refer to the source publication.

What WIFF predicts

WIFF represents wave-induced floe fracture within an evolving sea ice floe size distribution (FSD). The expensive SP-WIFF scheme generates a one-dimensional strain field from a wave spectrum, detects fracture lengths, and repeats stochastic sampling until the fracture histogram converges. NN-WIFF replaces this subroutine, rather than replacing the wave model, sea ice model, or complete prognostic FSD evolution. [Sections 1-2]

The learned target is a length-weighted fracture-length histogram. Together with the existing FSD, it determines which floes break and the distribution of daughter sizes. It is not simply a directly observed, instantaneous floe-size distribution. The present FSD is not a required direct input to the histogram network; it enters the subsequent FSD tendency calculation. [Equations 2-4]

Numerical target and physical bookkeeping

SP-WIFF constructs strain over an arbitrarily chosen 10 km domain and bins distances between relevant strain extrema. Sampling continues until the mean absolute change between successive histograms falls below 5 x 10^-4. This stochastic convergence is distinct from feedback convergence between coupled wave and ice models. [Section 2]

The fracture tendency removes area from larger floe categories and transfers it to smaller ones through a normalized daughter-size distribution. Mechanical breakup redistributes ice area across sizes rather than directly changing total ice concentration or thickness. Later dynamics and thermodynamics can change those bulk states.

A previous single-iteration, fixed-phase approximation, SP-WIFF1, reduces cost but introduces sampling error. The paper compares NN-WIFF with both the converged SP-WIFF reference and this cheaper numerical approximation, providing a useful accuracy-cost baseline.

Data, input contract, and network design

Six-hourly output from a coupled CICE-WW3 simulation for 2009 supplies both hemispheres. About 180 million candidate states include 17.9 million sea ice points; applying thickness below 10 m, concentration above 0.01, and significant wave height above 0.1 m leaves 5.1 million candidate network inputs. Approximately 2.6 million generate nonzero fracture histograms. These thresholds must be distinguished from the learned classifier. [Section 3]

Each input contains 25 wave spectral amplitudes, one ice thickness, and one ice concentration: 27 values in total. The output uses 12 floe categories. Spectral units, coordinate choice, integration weights, and normalization need to match the implementation; a lake spectrum with a different discretization is not a drop-in substitute.

The classifier has two hidden layers of 100 nodes each; the histogram network has five hidden layers of 100 nodes each. Both use fully connected feed-forward architectures, ReLU hidden activations, and softmax outputs. A separate gate is needed because a normalized softmax histogram cannot represent the all-zero no-fracture state. Data are randomly split 70/30 for training and validation. [Sections 3.1-3.2]

Losses, classification, and validation results

The histogram loss is representative size error (RSE): the sum of absolute category-density error multiplied by category radius and bin width. Standardized size error (SSE) divides RSE by the true representative radius. SSE here is not the conventional statistical sum of squared errors. A physically weighted metric is used because errors at different floe sizes have different consequences. [Equations 6-9]

The classifier threshold is selected on validation data, with a reported total classification error of 12.5% for the filtered candidate states. Figure 1 shows substantial dependence on the input regime, including poorly represented regions. The often-cited 96.5% overall call accuracy includes states eliminated by the preliminary energy cutoff and should not replace the filtered classifier's 87.5% accuracy. [Section 3.1 and Conclusions]

For roughly 800,000 histogram-validation states, median SSE is 3.9%. Large relative errors tend to occur for weak wave energy and large representative floe sizes, with problematic combinations of thin or low-concentration ice. Median category error is below 6.2% in every category and averages 3.3% across categories. These errors are against SP-WIFF-generated targets, not observed floe sizes. [Section 3.2; Figures 2-3]

Coupled-model behavior and computational cost

The independent-year evaluation uses CICE6.1.4 forced for 2005, with prescribed time-varying wave spectra, compared with the 2009 training conditions. NN-WIFF has median standardized size errors around 8-9% in the reported hemispheric comparison; the conclusions report an aggregate 8.4%. It performs similarly to or better than SP-WIFF1 in the selected comparison. [Section 4; Figure 4]

Sea ice area and volume differ little between NN-WIFF and converged SP-WIFF, and the paper's CICE quality-control test finds no emergent climate-changing difference for these integrations. Differences in representative floe size are more visible near the pack/MIZ transition where the classifier can miss fracture. This is evidence for the tested integration, not universal equivalence in every climate or region. [Figure 5]

On the stated Cheyenne configuration, one simulated year takes about 24 h with SP-WIFF, 3 h 30 min with SP-WIFF1, and 1 h 10 min with NN-WIFF. The FSD-without-waves reference takes 56 min. These are whole CICE integration times under the paper's settings; the wave model was run offline and its cost is excluded. They should not be advertised as standalone neural-inference timings or universal speedups. [Section 4]

Limits and a Great Lakes reproduction strategy

Training targets inherit the physical assumptions of SP-WIFF, including its simplified ice response and stochastic strain construction. A random split within one model year is useful for emulation accuracy but is weaker evidence of regional or climate transfer than independent-event, year, or model tests. The authors explicitly warn about out-of-distribution ice and wave states and provide retraining tools. [Section 5]

A sensible project sequence is to reproduce the published input contract, regenerate selected numerical targets, and verify histogram constraints and reported error definitions. Next, map Great Lakes spectra into the required bins with documented energy conservation, compare regional states with training coverage, and evaluate both a classifier and a histogram model. This is a proposed research plan, not a completed reproduction.

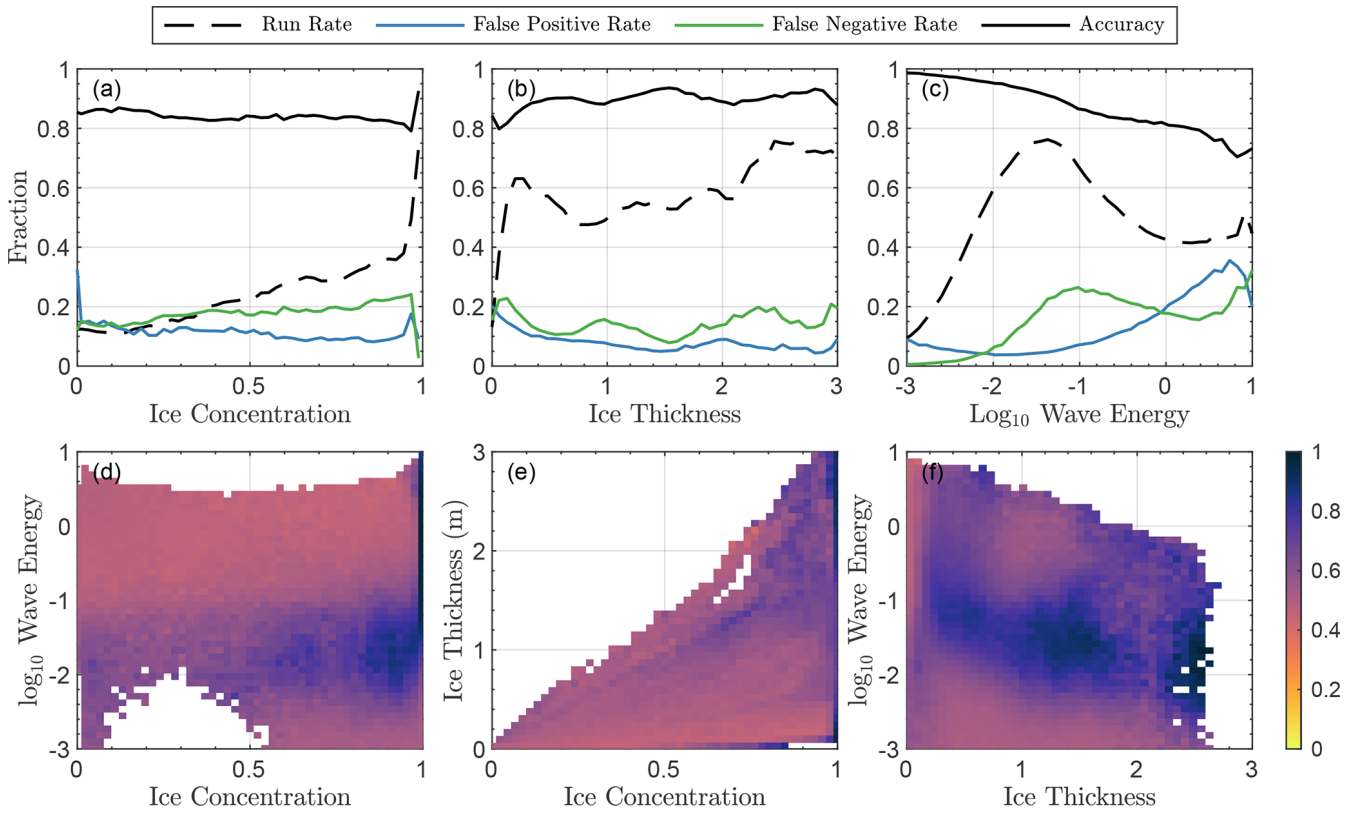
Code is archived at DOI 10.5281/zenodo.5793692; the paper identifies training data at 10.5281/zenodo.5108636 and model-output data at 10.5281/zenodo.5106703. These are the archival identifiers reported by the paper; present availability has not been checked here. An observational Great Lakes validation would require additional evidence beyond reproducing SP-WIFF labels.

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FIGURE ATLAS | 01 / 05

Figure 1. Classifier accuracy and run behavior



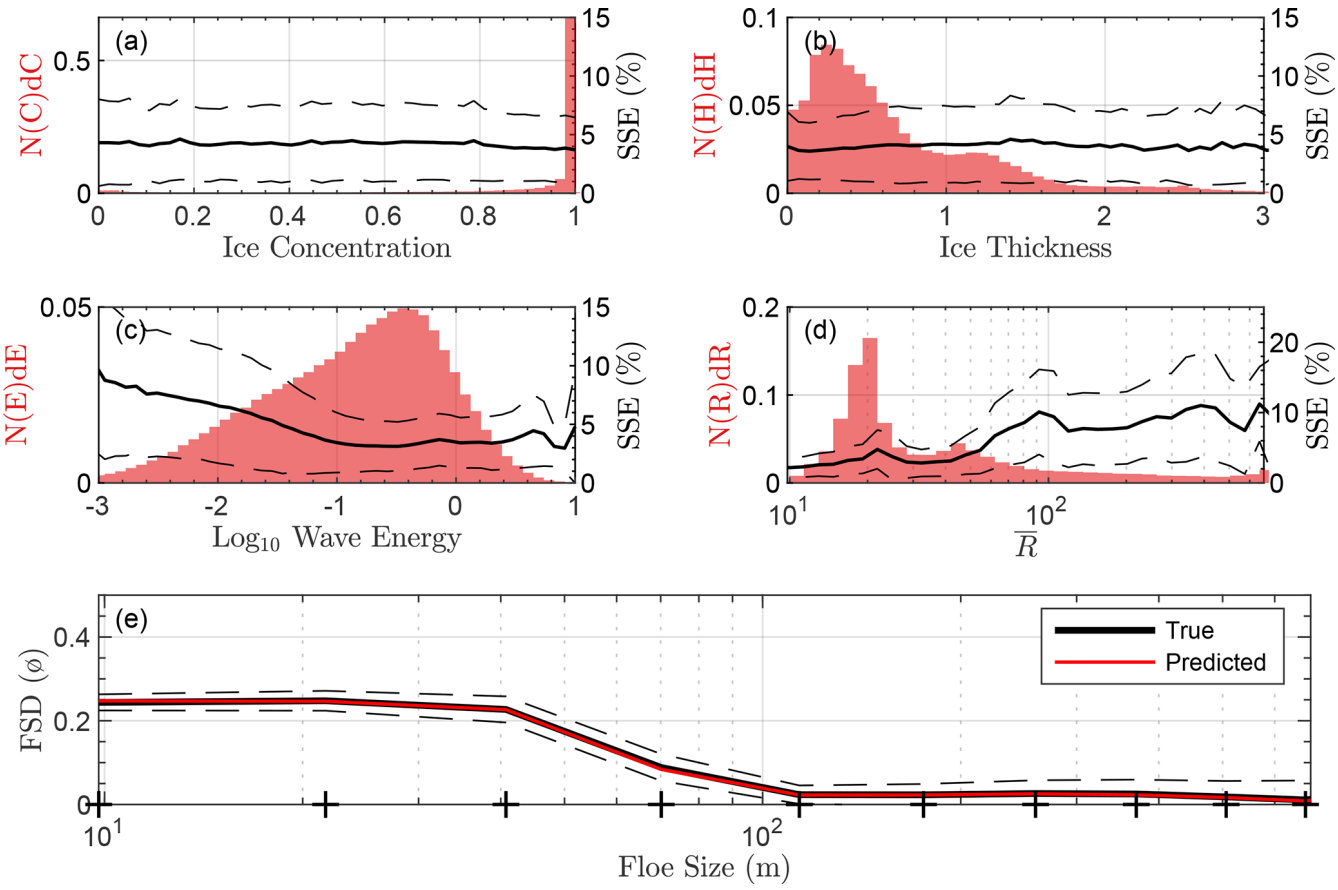
How to read this figure

Run rate, false positives, false negatives, and accuracy are shown across ice and wave conditions, together with two-dimensional accuracy maps. The classifier is a scientific component of the emulator, not just a computational convenience. Uncolored regions are under-sampled, and low-accuracy regions motivate targeted regional validation or retraining.

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FIGURE ATLAS | 02 / 05

Figure 2. Histogram validation and physical error



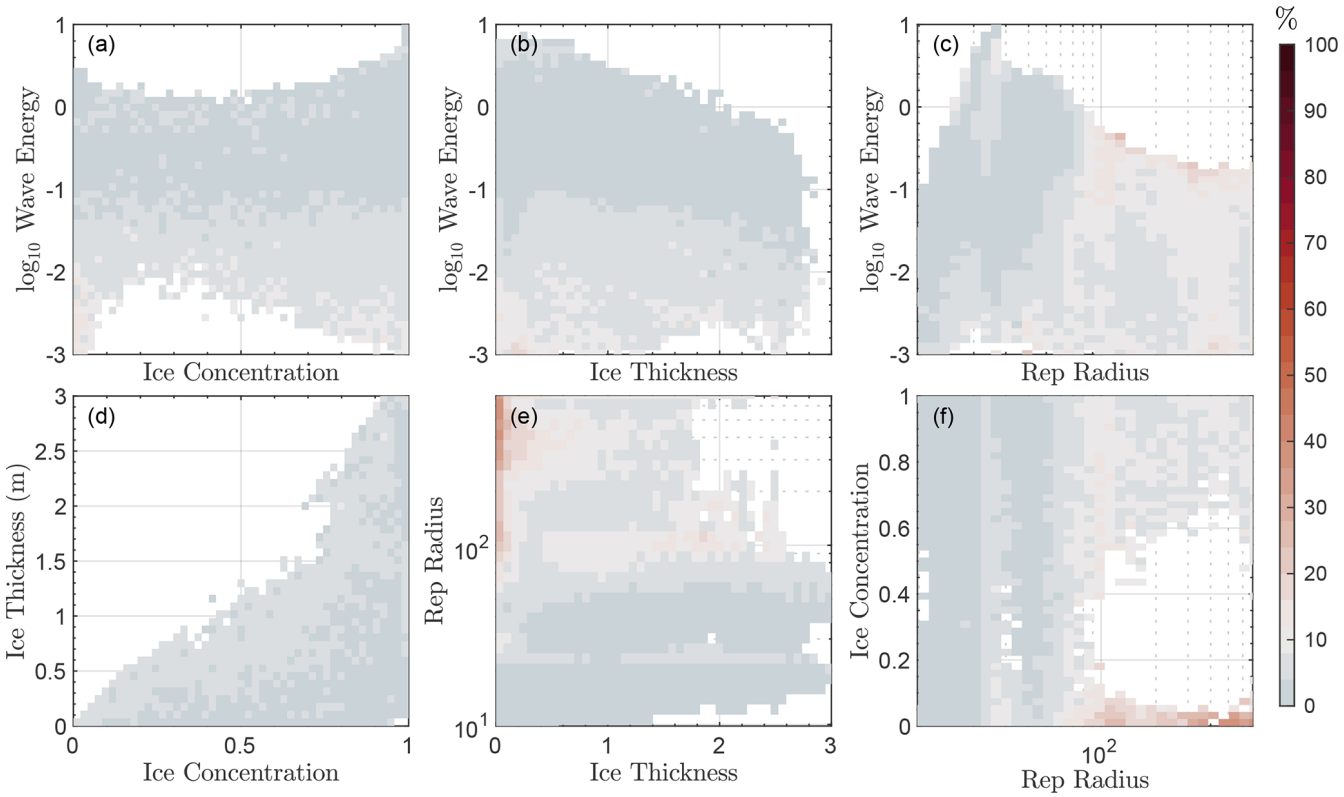
How to read this figure

Validation-state distributions are paired with standardized size error versus concentration, thickness, energy, and representative radius. The final panel compares mean predicted and reference fracture histograms across all twelve categories. The results quantify emulation of SP-WIFF statistics and expose where aggregate median error hides difficult states.

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FIGURE ATLAS | 03 / 05

Figure 3. Where histogram errors concentrate



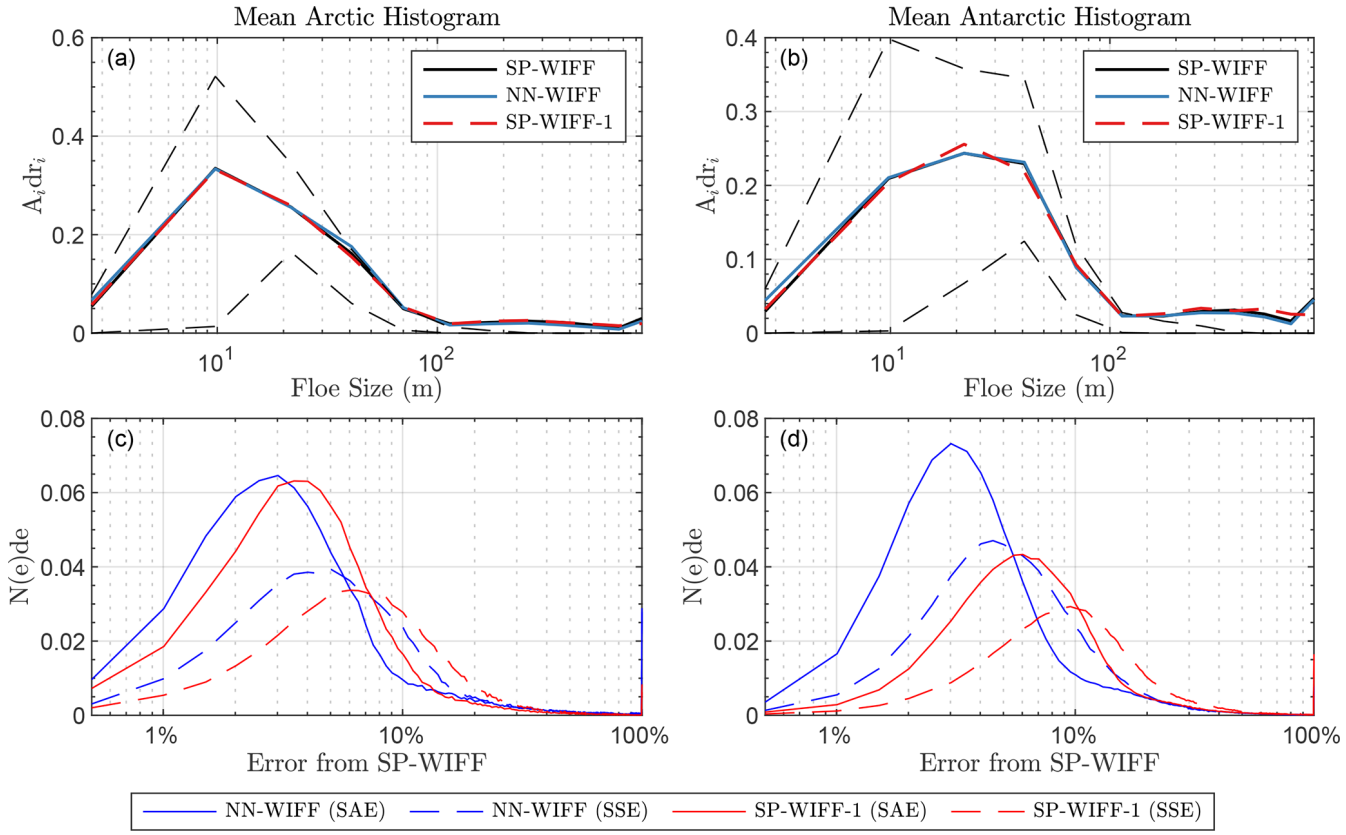
How to read this figure

Two-dimensional maps condition standardized size error on combinations of the physical inputs and representative radius. High error is associated with particular combinations rather than a uniform failure over all thicknesses or concentrations. Empty cells indicate limited coverage and should not be interpreted as good performance.

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FIGURE ATLAS | 04 / 05

Figure 4. Free-running CICE comparisons



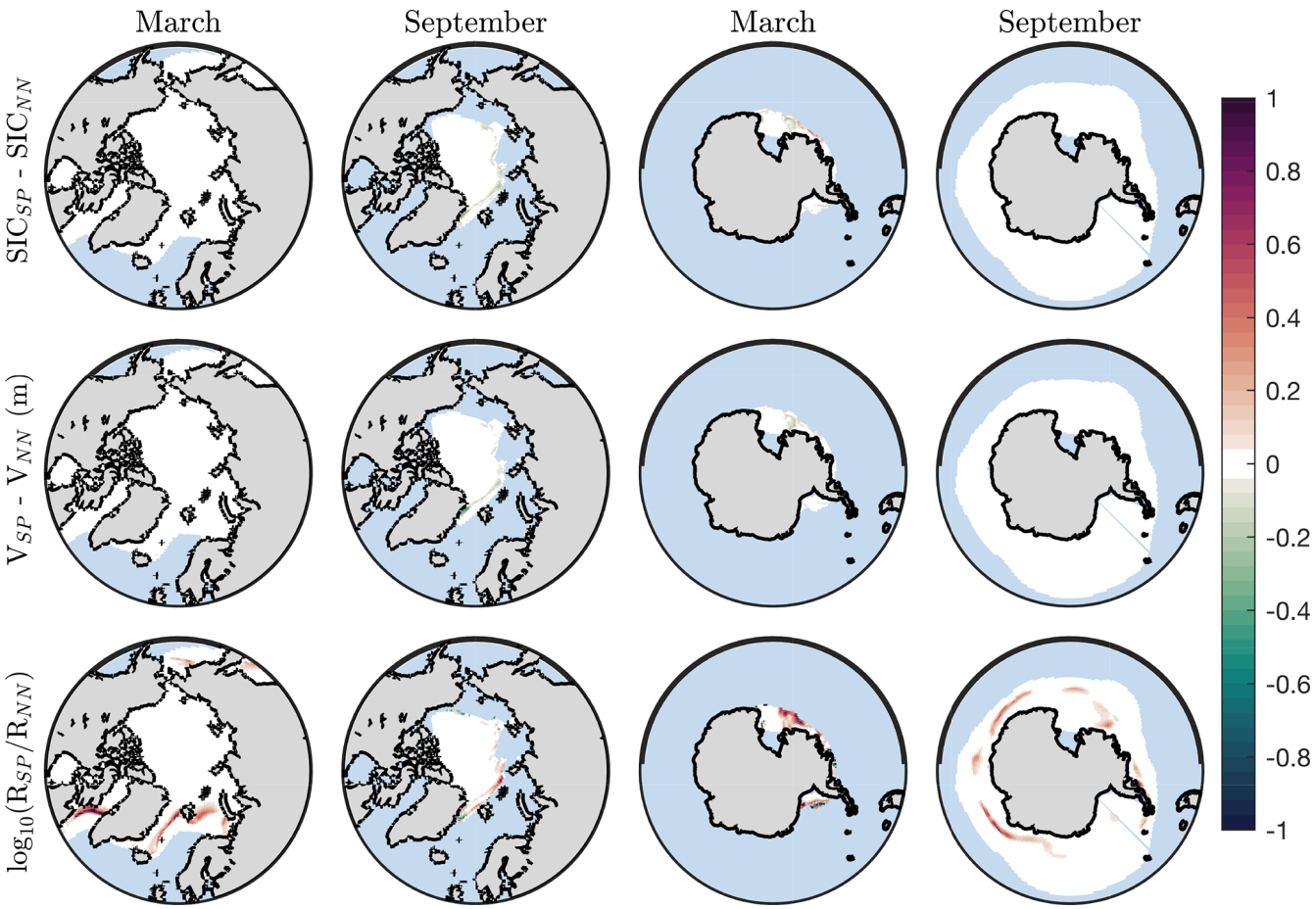
How to read this figure

Arctic and Antarctic mean fracture histograms and error distributions compare NN-WIFF and the single-iteration numerical approximation against converged SP-WIFF. The plotted sample and filtering conditions matter. Similar model statistics support acceleration in this setting but do not constitute an independent field validation of the fracture parameterization.

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FIGURE ATLAS | 05 / 05

Figure 5. Emergent ice-state differences



How to read this figure

Maps compare ice concentration, volume per area, and representative floe radius between SP-WIFF and NN-WIFF for March and September. Bulk sea ice states are similar, while floe-size differences can appear near the transition into pack ice. This final test connects subroutine error to behavior of the larger sea ice model.

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